

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |

Kent, OH - F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) with a track record of delivering complete AI systems end-to-end - from stakeholder requirements to deployed, containerized production services. Shipped healthcare RAG pipelines (~35% retrieval precision gain), multi-agent reasoning systems, and agentic LLM workflows that improve measurable outcomes. Fast learner and force-multiplier: independently owned ML and AI delivery with no dedicated ops team. Experience leading teams of 150+ and formalizing onboarding. US-based on F-1 OPT, seeking sponsorship.

CORE COMPETENCIES

End-to-End AI Delivery • Agentic LLM Systems • RAG & Production ML • Python / FastAPI / Docker • TypeScript / React • Client Requirement Translation • Multi-Workstream Management • AI Tools Daily Fluency • Team Mentorship • LangChain / LlamaIndex • Healthcare AI (HIPAA) • Fast Prototyping

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS

- Built **production RAG systems** for clinical knowledge retrieval with recursive semantic chunking and transformer embeddings; improved contextual retrieval precision ~**35%**, reduced irrelevant retrieval >**30%**, retrieval-grounded response alignment exceeded **90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning and tool discipline; improved agent response stability ~**25%** and reduced hallucinated outputs >**30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall **15-20%** on high-risk categories at >90% precision.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services with Docker, structured logging, and load simulation; reduced post-deployment defects ~**30%**.
- Maintained HIPAA-conscious data governance with de-identification, data lineage, and evaluation audit trails.

Kent State University

Aug 2023 - May 2025

Student Manager - Operations, Team Leadership

- Led coordination for **150+ students**; reduced coordination delays ~**30%** and formalized onboarding cutting ramp-up ~**20%**.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP

- Optimized T-SQL stored procedures for Order-to-Cash platform; reduced query latency ~**25%** and execution time ~**20%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and SQL deployments; reduced deployment errors >**30%**, improved release cycle ~**35-40%**.
- Designed Grafana dashboards tracking long-running queries, deadlocks, and I/O contention; cut incident recurrence ~**25%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics

- Built transformer-based video summarization system across 5,000+ sessions; **60-70% reduction** in manual review time, ~**85% alignment** with human-curated highlights.
- Implemented document Q&A with RAG; reduced hallucinated outputs ~**30%**, raised contextual accuracy >**85%**.

FEATURED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate detection; explainable risk scoring with audit-ready traces.

Python, LangChain, LlamaIndex, FastAPI, Docker, RAG, agentic orchestration

Manga Lens - Chrome Extension for Real-Time AI Translation shipped, Chrome Web Store

Full-stack Manifest V3 extension integrating four AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-provider webhook-style routing with per-domain selector configs for 29 sites, seven-day translation cache, and viewport-slice screenshot pipeline with coordinate remapping.

TypeScript, React, Manifest V3, multi-provider API orchestration, WebP/JPEG routing

Dream Decoder - Multimodal AI Creative Platform

FastAPI + React/TypeScript/Vite/Tailwind app coordinating multiple AI APIs (Perplexity Sonar, Replicate image models). Intermediate structured prompt transformation layers improved contextual alignment ~30% and first-pass image success rate ~25-30% over naive direct-prompt approaches.

FastAPI, React, TypeScript, multi-LLM orchestration, structured outputs, REST APIs

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN with unified YOLOv8; reduced inference latency ~30%. Sliding-window confidence aggregation suppressed blink-driven false positives ~25%; adaptive frame skipping and NMS tuning for stable real-time OpenCV capture.

Python, YOLOv8, OpenCV, PyTorch, LabelImg, real-time inference

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages: Python, TypeScript, JavaScript, FastAPI, Flask, React, SQL (T-SQL, PostgreSQL)

AI & LLM: RAG, agentic workflows, multi-agent orchestration, structured outputs, evaluation pipelines

ML Libraries: LangChain, LlamaIndex, scikit-learn, XGBoost, Hugging Face Transformers

Infrastructure: Docker, FastAPI services, Jenkins CI/CD, Grafana observability, RESTful APIs

Domains: Healthcare AI, Enterprise AI, Workflow Automation, Applied ML